Genomic Features for Recurrence Risk Stratification in METABRIC With an Exploratory TCGA Assessment

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Abstract

Background. This research evaluated whether gene-level molecular summaries improve recurrence discrimination in METABRIC and whether the observed signal transfers to TCGA-BRCA.

Methods. The complete-case METABRIC cohort included 1,295 patients with 545 recurrences, including 996 ER-positive patients with 408 events. Gene-expression, copy-number and mutation features were added to a legacy baseline containing clinical, treatment and mutation-count variables, and to a strictly clinical and treatment comparator. Fixed-penalty and tuned survival models were evaluated using 25 and 50 repeated cross-validation splits, respectively, on previously cohort-normalized expression data. An exploratory TCGA assessment used refitted models with six harmonized predictors and within-cohort molecular standardization.

Results. In ER-positive METABRIC patients, tuned Cox discrimination increased from 0.632 to 0.658 (difference 0.026; corrected 95% confidence interval −0.002 to 0.054). A reduced-feature random survival forest increased from 0.629 to 0.662 (difference 0.033; 95% confidence interval 0.015–0.051; within-family Holm-adjusted p = 0.002), and its gain over the strictly clinical comparator was 0.043 (0.024–0.061). These model-development analyses were exploratory. In 679 TCGA patients classified as ER-positive from ESR1 expression, with 66 progression events, the harmonized Cox gain was 0.014 (95% bootstrap confidence interval −0.036 to 0.064) and −0.009 with censoring weighting at 36 months. Gains were not demonstrated in the combined METABRIC cohort or the five-year landmark analysis.

Conclusions. Genomic summaries showed an internal signal of improved discrimination in ER-positive METABRIC patients. External incremental performance remains uncertain because of imprecision, endpoint differences and cohort adaptation. Independent evaluation of a fully specified preprocessing and prediction pipeline, including calibration, is needed before clinical interpretation.

Keywords: breast cancer; recurrence; METABRIC; TCGA-BRCA; multi-omics; survival analysis; random survival forest; transportability

1. Introduction

A recurrence is the regrowth of tumor cells that had already disseminated before or around the time of treatment. Clinical features such as tumor size, nodal status, and the Nottingham Prognostic Index (NPI) measure how much disease had the opportunity to spread. They describe the intrinsic behavior of the tumor

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cells only coarsely, through histologic grade and three receptor markers. Gene-expression assays measure that behavior more directly. Several of them, including the 21-gene Recurrence Score, PAM50 risk of recurrence, EndoPredict, and the Breast Cancer Index, add prognostic information to clinicopathological variables, particularly in ER-positive disease [2–4]. Expression modules associated with outcome also differ by subtype: proliferation dominates in ER+/HER2− tumors and immune response in ER−/HER2− tumors [5].

ER-positive disease is also where clinical need is greatest over the long term. Recurrences continue at a steady rate for at least 20 years after five years of endocrine therapy [6], and a 2025 patient-level meta-analysis showed that extending aromatase-inhibitor therapy beyond five years reduces recurrence by about a quarter at the cost of side effects, so better risk stratification could inform such decisions [7]. METABRIC is the cohort in which the Integrative Clusters were defined [8] and in which copy-number-driven ER-positive subgroups with high late-relapse risk were described [9]. Whole-genome work has since shown that high-risk ER-positive tumors carry complex focal amplifications and structural rearrangements, while typical-risk ER-positive tumors are largely genomically stable [10].

At the same time, benchmark studies caution that multi-omic data often add little to a well-specified clinical model. Across TCGA cancer types, few multi-omic survival methods beat a clinical Cox model, and only slightly [11]; many multi-omic models also degrade when uninformative features are added [12]. Recent large-scale prognostic models in breast cancer that do improve clinical variables have relied on external validation across many cohorts [13]. These findings set two expectations for our design: gains from genomic data should be **modest** (on the order of 0.02 to 0.05 in discrimination), and they should be **concentrated in ER-positive disease**.

In our earlier work, recurrence was the hardest of several METABRIC targets, with a mean AUC of 0.666 from clinical and treatment features [1]. A pilot study (Phase 1) then tested the categorical molecular labels already present in the clinical table (PAM50, IntClust, 3-gene classifier, HER2 copy-number state, cellularity). They added only +0.004 to +0.008 AUC, and the labels largely duplicated ER, HER2, and grade information. The present study moves to gene-level data, uses time-to-event models, and adds an exploratory external assessment; it reuses METABRIC, so its internal results are not independent of the EGETC study. It asks three questions: (1) do expression, copy-number, mutation, methylation, and immune features improve recurrence discrimination over the EGETC baseline and over a strictly clinical comparator; (2) is any gain concentrated in ER-positive disease or in late recurrence; and (3) is a similar signal seen in TCGA-BRCA?

2. Methods

Reporting draws on the TRIPOD+AI guidance for prediction models that use regression or machine learning [14]; a completed checklist is not claimed.

2.1 Data sources and cohort

All molecular data were obtained from the cBioPortal *brca_metabric* study [8,15–18] (Table 1). All files share the METABRIC patient identifier (MB-xxxx); each patient has one tumor sample whose identifier equals the patient's identifier (verified for all 2,509 patients). The final dataset has one row per patient and 141 columns, and a feature dictionary documents every engineered column.

Table 1. Data sources.

Source	Content	Patients
METABRIC clinical table (as in the EGETC paper)	34 clinical, pathology, treatment and outcome columns	2,509
mRNA z-scores (Illumina HT-12; provider-normalized)	20,603 genes	1,980
Copy number (GISTIC, −2 to +2)	22,544 genes	1,985
Mutations (173-gene targeted panel)	17,272 mutations	2,433 sequenced
Promoter methylation (RRBS, beta 0–1)	13,188 promoters	1,418

Outcome. Relapse-free status (recurred vs not recurred) and relapse-free months from diagnosis, as recorded in cBioPortal. Patients without a recorded relapse were censored at their last relapse-free follow-up; the endpoint does not distinguish local from distant relapse, and deaths without relapse were not modelled as competing events. Unknown outcomes were left missing and never treated as negative.

Modelling cohorts. Of the 2,509 patients, 2,380 had a recorded relapse-free status and time, 2,376 had follow-up above zero, and 1,295 had complete legacy baseline variables; all 1,295 also had expression, copy-number, and mutation data (545 recurrences; 996 ER-positive with 408 recurrences), i.e. 51.6% of the clinical table. Compared with the 1,081 patients who had a recorded outcome but were excluded, included patients more often recurred (42.1% vs 38.5%), were ER-positive (76.9% vs 69.0%), grade 3 (52.4% vs 42.1%), node-positive (46.9% vs 40.2%), and treated with chemotherapy (21.9% vs 11.7%), and had longer follow-up (163 vs 147 months by reverse Kaplan–Meier). Methylation analyses used the 945 of these patients with all 14 modelled methylation features. Cohort characteristics are shown in Table 2.

Table 2. Characteristics of the modelling cohort.

Characteristic	All (n = 1,295)	ER-positive (n = 996)	ER-negative (n = 299)
Recurrences, n (%)	545 (42.1)	408 (41.0)	137 (45.8)
Age at diagnosis, median (IQR), years	61.1 (50.8–69.7)	62.9 (53.0–70.6)	54.4 (44.1–65.1)
HER2-positive, %	12.4	7.2	29.4
Lymph-node positive, %	46.9	45.2	52.5
Histologic grade 3, %	52.4	41.7	88.3
Tumor size, median (IQR), mm	22 (17–30)	22 (17–30)	25 (16.5–31.5)
Chemotherapy / hormone therapy / radiotherapy, %	21.9 / 60.5 / 66.9	12.0 / 70.9 / 64.9	54.5 / 25.8 / 73.9
Censored without recurrence before 60 months, n	88	73	15
Median follow-up (reverse Kaplan–Meier), months	163	159	174

2.2 Feature engineering

Molecular features were computed per patient from that patient's own measurements and fixed, published gene lists, without outcome information. The expression inputs, however, were the cBioPortal mRNA z-scores (file data_mrna_illumina_microarray_zscores_ref_diploid_samples, downloaded in May 2025 and archived with the code), which the data provider standardized against a reference population drawn from the METABRIC

cohort. This upstream normalization was not re-estimated inside cross-validation folds, so all internal evaluations use previously cohort-normalized data and are not strictly inductive.

Baselines. The legacy baseline of the EGETC paper [1] had 22 columns: age, grade, positive nodes, mutation count, NPI, tumor size and stage, ER/PR/HER2 status, chemotherapy, hormone therapy, radiotherapy, mastectomy, menopausal status, and derived indicators (node-positive flag, high grade, log mutation count, size in cm, NPI group, age group, ER×HER2 interaction). It is not purely clinical: it includes treatment received and two mutation-count summaries, and log mutation count correlates 0.999 with the mutation-count feature of the genomic block. Constant and exactly duplicated columns were removed within each cohort (size in cm duplicates tumor size; in ER-positive patients ER status is constant and ER×HER2 equals HER2 status), giving the baseline used throughout (BASE: 21 columns in all patients, 19 in ER-positive patients); on identical splits this cleaning changed the Cox C-index by less than 0.001. A strictly clinical and treatment comparator (CLIN) also removed mutation count and log mutation count (19 and 17 columns). NPI was kept alongside its components, and continuous variables alongside their categories, to preserve the published specification; these columns are partly redundant.

Gene-expression gene-set scores (57 features). With about 1,300 patients and 20,000 genes, modelling individual genes invites overfitting and produces results that are hard to interpret. Gene sets group genes that act together in a biological process, compress the data, and make model explanations meaningful. We used all 50 MSigDB Hallmark sets [19,20] (human collection, release 2026.1, h.all.v2026.1.Hs) without hand-picking; each score is the mean z-score of the member genes present in METABRIC (95–100% coverage, median 99%). We also computed the seven Desmedt breast-cancer modules [5], built around prototype genes AURKA (proliferation), ESR1 (ER signaling), ERBB2 (HER2 signaling), STAT1 (immune response), PLAU (invasion), VEGF (angiogenesis), and CASP3 (apoptosis). Each module score is the weighted average of member-gene z-scores using the published coefficients:

$$S_m = \frac{\sum_i w_i z_i}{\sum_i |w_i|}$$

where w_i is the published coefficient and z_i the expression z-score of gene i in module m, with genes matched by Entrez ID (905 of 913 found). Duplicate probes for the same gene were averaged.

Copy-number driver features (11). High-level amplification (GISTIC +2) of nine recurrently amplified breast-cancer drivers, several of which define Integrative Clusters linked to late relapse [8,9]: CCND1 (16.5% of tumors), PAK1 (9.1%), FGFR1 (12.9%), ZNF703 (14.0%), MYC (25.2%), RPS6KB1 (10.3%), PPM1D (9.9%), ERBB2 (15.1%), and MDM2 (3.2%). Deep deletions of CDKN2A, PTEN, RB1, and MAP2K4 are each rare, so they were combined into one indicator (6.8% of tumors). Missing gene-level calls were kept missing (none occurred at these 13 genes). A copy-number instability proxy, the fraction of assayed genes with a non-zero GISTIC call, was added; missing calls (2,356 calls in 82 tumors, at most 0.35% of genes per tumor) were excluded from numerator and denominator. This gene-count fraction is not weighted by genomic length and is therefore not the fraction of the genome altered.

Mutation driver features (9). An indicator of any protein-changing (non-silent) mutation in eight frequently mutated drivers [18]: PIK3CA (40.1% of sequenced tumors), TP53 (35.5%), KMT2C (11.5%), GATA3 (11.1%), MAP3K1 (9.7%), CDH1 (9.7%), AKT1 (4.1%), and PTEN (3.9%), plus the log count of non-silent variants on the

173-gene panel (a variant count, not tumor mutational burden per megabase). Silent, intronic, UTR, and flanking variants were excluded. Sequenced patients without a mutation in a gene were coded 0.

DNA methylation features (20 built, 14 modelled). Motivated by the finding that epigenomic instability in METABRIC tumors correlates with grade, stage, TP53 mutation, and poorer prognosis [21]. Features were global mean promoter methylation, within-tumor spread, fraction of strongly methylated promoters (beta > 0.5), fraction of intermediate values (0.2–0.8) as an epigenomic-disorder proxy, mean promoter methylation of each Desmedt module's genes, and promoter methylation of nine breast-cancer genes. The cBioPortal file contains no normal-tissue samples, so instability was measured within each tumor rather than against normal breast. Single-gene promoters missing in more than 1% of methylation-profiled tumors (for example BRCA1 and ESR1) were excluded, leaving 14 features. This rule uses predictor missingness only; it was applied once to all profiled tumors before cross-validation, and the resulting list was frozen.

Immune and stromal marker summaries (10). Immune composition is prognostic in breast cancer, with regulatory T cells and M0/M2 macrophages linked to poor outcome and CD8+ T cells to better outcome in ER-negative disease [22], and tumor-infiltrating lymphocytes strongly prognostic in triple-negative disease [23]. MCP-counter marker genes [24] were used to compute marker-based z-score summaries for ten populations: T cells, CD8 T cells, cytotoxic lymphocytes, NK cells, B lineage, monocytic lineage, myeloid dendritic cells, neutrophils, endothelial cells, and fibroblasts. Each summary is the mean expression z-score of the population's marker genes; this adapts the MCP-counter signatures but is not the original MCP-counter abundance estimator, and the values are not cell proportions.

2.3 Redundancy analysis

Many Hallmark sets overlap, and correlated features can split importance between them, inflate model complexity, and mislead interpretation. We measured this redundancy with principal component analysis (PCA) of the 57 standardized gene-set scores across all 1,980 profiled tumors and with average-linkage hierarchical clustering of $1 - |r|$, cut at 0.3 (this does not guarantee $|r| \geq 0.7$ for every pair within a cluster). PCA does not use the outcome, so it describes structure, not predictive value. When principal components were used as model features, PCA was refitted inside each training fold and inside each inner fold during tuning. The number of components (9, reaching 80% of variance in this descriptive analysis) was a fixed design choice taken from this analysis, not tuned.

2.4 Endpoints and feature sets

Endpoints and analysis hierarchy. The primary endpoint analysis is time to recurrence, which uses follow-up and censoring, scored by Harrell's C-index [25]. The primary question was the increment from genomic features in ER-positive patients; all patients and a five-year landmark analysis (patients recurrence-free and under follow-up at 60 months, time measured from month 60, event any recorded relapse, baseline information only) were secondary. Binary classification of recorded Recurred versus Not Recurred status (logistic regression and LightGBM, ROC AUC) ignores follow-up and censoring—88 patients without a recorded event, 73 of them ER-positive, were censored before 60 months—so it is reported only as descriptive classification of observed status, for comparability with the EGETC paper. The tuned models, the reduced representation (PC+DRV), the strictly clinical comparator, and the CORE4 set were added after an earlier round of analyses on the same data (CORE4 after inspecting SHAP results); these analyses are exploratory.

Prediction time. The baseline includes treatment received (chemotherapy, hormone therapy, radiotherapy, mastectomy), which is known only after surgery and treatment planning. The models therefore describe prognosis after primary treatment planning; they are not a pretreatment decision tool, and treatment coefficients are not estimates of treatment benefit.

Feature sets. BASE (21 columns in all patients; 19 in ER-positive patients); BASE+DM (+7 Desmedt modules); BASE+HM (+50 Hallmark sets); BASE+PC (+9 principal components of the 57 scores); BASE+DRIVERS (+11 copy-number and 9 mutation features); BASE+GEN (+all 77 genomic features); BASE+PC+DRV (reduced representation: 9 principal components + 20 drivers); BASE+SHAPSEL (+10 genomic features selected by SHAP inside each training fold); BASE+CORE4 (AURKA proliferation, ESR1 signaling, STAT1 immune, and the copy-number fraction); the same genomic blocks added to CLIN; and BASE+METH (+14), BASE+IMM (+10), and their combinations with GEN. Every comparison within an analysis uses the same patients and the same splits.

2.5 Models

- **Penalized (ridge) Cox regression** (scikit-survival [26]). In the 25-split analyses, the penalty was fixed at $\alpha = 10$; in the tuned analysis it was chosen from {1, 10, 100, 1000} by inner 3-fold cross-validated C-index, with scaling, PCA, and model fitting repeated inside each inner training fold.
- **Random survival forest (RSF)** [27]: 150 trees, $\sqrt{p}$ features per split, minimum leaf size chosen from 15 or 40 by inner 3-fold cross-validated C-index of the whole pipeline.
- **L2 logistic regression** ($C = 0.1$) and **LightGBM** [28] (300 trees, learning rate 0.03, 15 leaves, minimum 30 samples per leaf, row and column subsampling) with fixed settings, for descriptive classification of observed status and for SHAP.
- Earlier exploratory AutoML classification of observed status (FLAML [29] and MLJAR [30], single 5-fold split, Nadeau–Bengio test [32]) used the previous predictor specification and is not reported; its code and archived results are in the repository.

2.6 Validation and statistical testing

We used stratified 5-fold cross-validation repeated 5 times (25 splits) for the fixed-penalty models and 10 times (50 splits) for the tuned survival models. Scaling, PCA, SHAP-based feature selection, and hyperparameter selection were fitted inside the training fold (and, for tuning, inside inner folds); the upstream expression normalization was not (Section 2.2). Paired differences from the matching baseline were tested with the corrected repeated k-fold t-test [31], with Holm adjustment [33] within each declared family of comparisons (stated with each table); Holm adjustment does not cover the adaptive decisions made across the project. Correlated test folds make the ordinary paired t-test anticonservative, and cross-validation estimates average performance over training sets rather than the error of one fitted model [34]; repeated splits are not independent replications. As a censoring-sensitive check, Uno's inverse-probability-of-censoring-weighted C-index [43] was recomputed from the saved out-of-fold predictions, truncated at 120 months, with the censoring distribution estimated in each training fold. Seed 42 was used for splitting and models, and seed 0 for the ESR1 mixture model.

2.7 Interpretability

TreeSHAP [35] was computed for the binary LightGBM classifier on BASE+GEN in each of the 25 test folds (log-odds scale); we report mean $|\text{SHAP}|$ and how often each feature ranked in the top 15. These values describe that classifier in all patients; they do not explain the ER-positive survival-model results, they split credit among

correlated features, and they are model-dependent predictive attributions, not causal or biomarker evidence. The top 10 genomic features were also selected by SHAP inside each training fold and evaluated on the held-out fold (BASE+SHAPSEL).

2.8 Exploratory assessment in TCGA-BRCA

All results above are internal to METABRIC, a single historical UK/Canadian cohort. TCGA-BRCA [36] (PanCancer Atlas release on cBioPortal; US patients diagnosed mostly in the 2000s–2010s) was used for an exploratory transport assessment [37]. The same feature formulas were applied, but measurement differs: RNA sequencing instead of microarrays, and whole-exome instead of 173-gene panel sequencing, so the mutation count covers a different gene universe (a sensitivity analysis excluded it). Every TCGA expression, copy-number, and mutation sample was a primary tumor (sample type 01), with one sample per patient in each data type.

TCGA does not provide histologic grade, NPI, node counts, chemotherapy or hormone-therapy flags, or receptor immunohistochemistry in this release, so a separate harmonized model was specified with six predictors (HCLIN): age; T category (METABRIC approximated from tumor size: ≤20, ≤50, >50 mm; TCGA from pathological T, with T4 recoded as T3); N category (METABRIC approximated from positive-node counts: 0, 1–3, 4–9, ≥10; TCGA from pathological N); ER status (METABRIC immunohistochemistry; TCGA inferred by a two-component Gaussian mixture on log2 ESR1 RNA-seq expression, which called 75.5% of tumors and 99% of PAM50 luminal tumors ER-positive, with no compatible clinical ER measurement available to validate it); ERBB2 high-level amplification as a proxy for clinical HER2; and radiotherapy. The influence of the T4 recoding and of these proxies was not assessed. Because ERBB2 amplification is in HCLIN, the genomic block has 76 features here. Models were refitted on METABRIC with these predictors rather than the internal baseline, tuned with the same inner cross-validation, and applied once to TCGA. Continuous molecular features were standardized within each cohort, using the whole unlabeled TCGA cohort to estimate TCGA means and standard deviations. This unsupervised target-cohort adaptation, together with the ESR1 mixture fitted to TCGA, means the assessment does not evaluate a frozen predictor that could be applied to a single new patient.

The TCGA outcome was the progression-free interval (PFI), recommended by the PanCancer Clinical Data Resource because it is defined for nearly all patients and accrues more events than overall survival over short follow-up [38]; time is measured from diagnosis, events follow the Resource's definition (including local or distant recurrence, new primary tumor, and progression), and other patients were censored at last follow-up. METABRIC relapse-free survival (Section 2.1) is defined differently. Differences in endpoint definitions, follow-up and case mix limit direct comparison and may affect the estimated incremental discrimination in either direction. Male patients (12), patients with M1 disease (20), and patients with neoadjuvant treatment (5) were excluded; 158 retained patients had unknown M stage (MX) and one had unknown neoadjuvant status, so retained patients are not all confirmed non-metastatic or untreated. Uncertainty was estimated with 1,000 paired bootstrap resamples of the TCGA set, and Uno's C-index truncated at 36 months was computed as a censoring-sensitive check.

3. Results

Figure 1 summarizes the increments. Genomic features improved internal discrimination in ER-positive patients, most clearly for the tuned reduced-feature random survival forest; improvement was not demonstrated in all patients or in the landmark analysis; and in TCGA all gain intervals included zero.

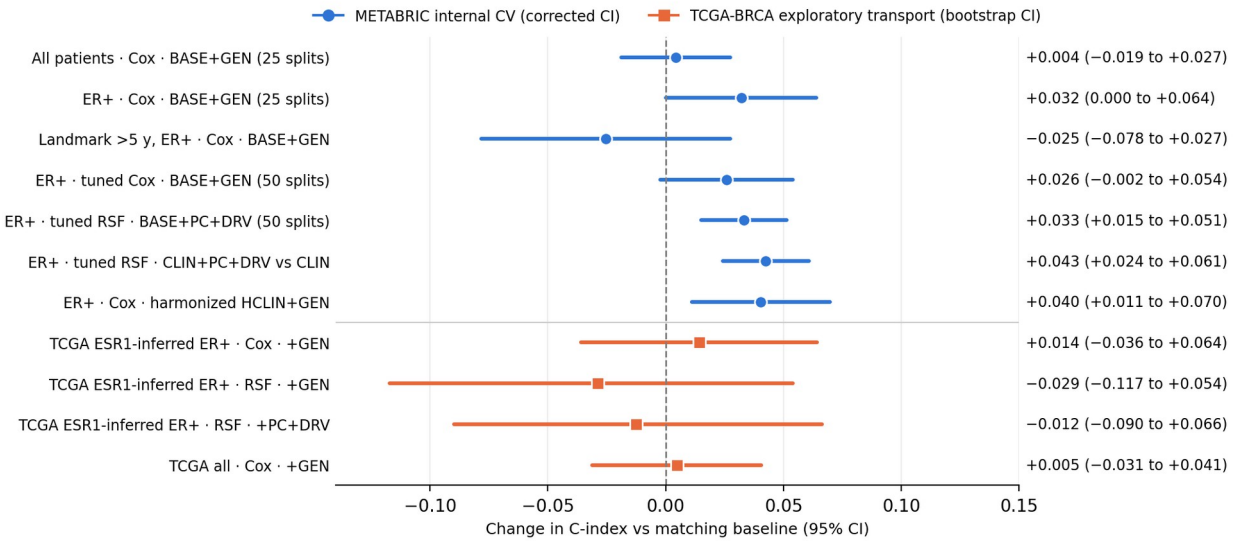


Figure 1. Change in C-index from adding genomic features to the matching baseline (BASE, CLIN, or HCLIN). Blue circles: internal METABRIC cross-validation (corrected 95% CI). Orange squares: exploratory TCGA-BRCA assessment of refitted harmonized models (bootstrap 95% CI).

3.1 Biological checks of the engineered features

The engineered features behaved as expected. The Desmedt proliferation score rose with grade (−0.46, −0.20, and +0.27 for grades 1–3) and correlated 0.90 with the Hallmark E2F-targets score. The ER-signaling score was higher in ER-positive tumors (+0.21 vs −0.68). ERBB2 amplification was present in 95.1% of HER2-positive tumors and 3.6% of HER2-negative tumors. CCND1 amplification was found in 79% of IntClust 2 tumors, and TP53 mutation was most frequent in basal tumors (88%). PGR promoter methylation was higher in PR-negative tumors, BRCA1 promoter methylation was highest in basal tumors, and CD8 T-cell scores were highest in claudin-low and basal tumors.

3.2 Redundancy among gene-set scores

Across 1,980 tumors, the 57 scores compressed to 9 principal components for 80% of the variance (18 for 90%). PC1 (24.5%) was an immune and inflammation axis, PC2 (20.3%) a proliferation axis, and PC3 (13.8%) a metabolic and adipose axis. Hierarchical clustering grouped the 57 scores into 32 clusters; the largest were inflammation and immune signaling (8 sets), proliferation (7 sets, including Desmedt AURKA, E2F targets, G2M checkpoint, and MYC targets; the mitotic spindle score formed a separate cluster), invasion and EMT (4 sets, including Desmedt PLAU), and ER signaling (3 sets, including Desmedt ESR1 and both estrogen-response Hallmarks) (Figure 2).

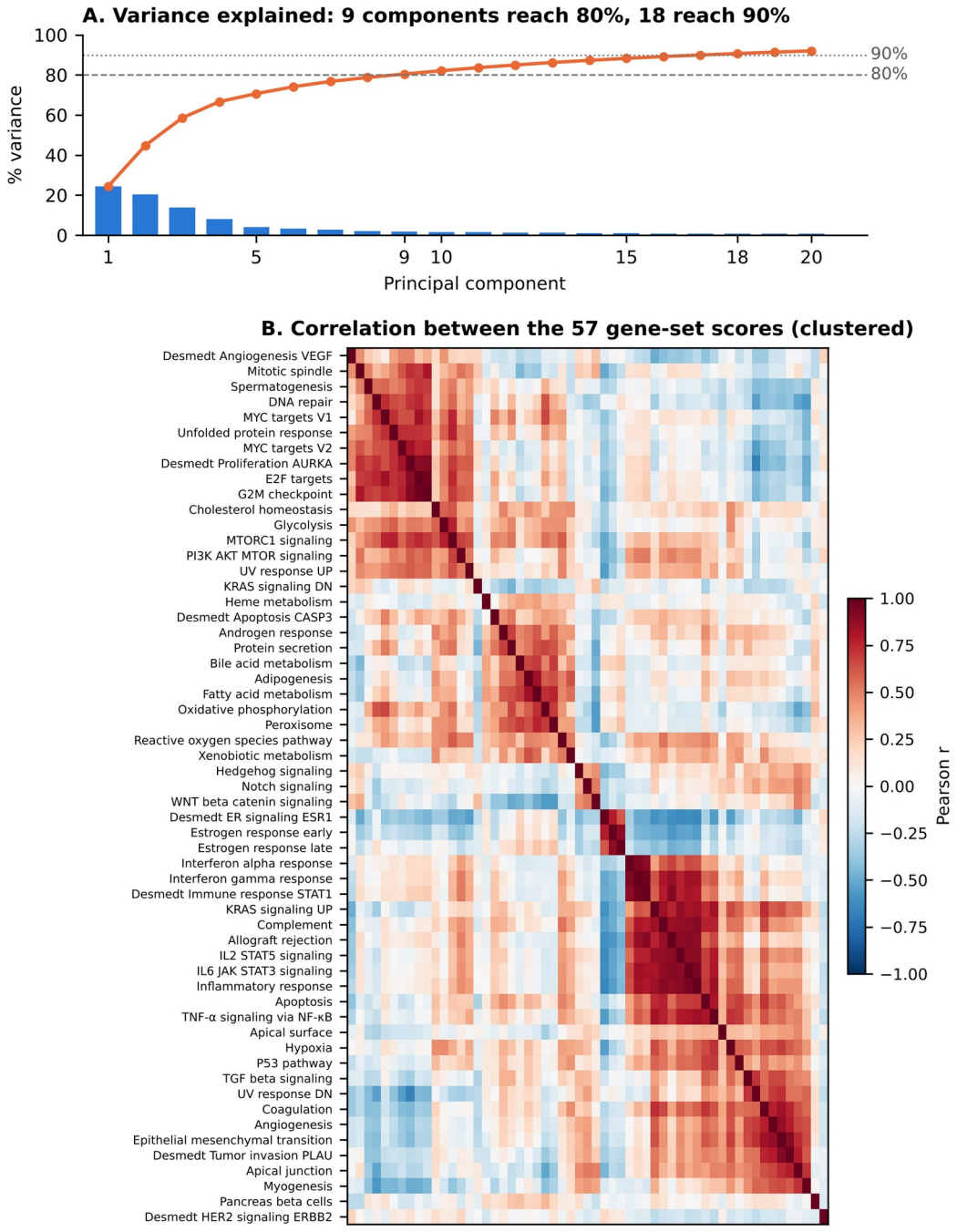


Figure 2. Redundancy among the 57 gene-set scores (1,980 tumors). A: variance explained by principal components (bars) and cumulative variance (line); 9 components reach 80%. B: correlation matrix of the scores, ordered by hierarchical clustering.

3.3 All patients: feature-set comparison

No feature set significantly improved BASE for all patients combined (Table 3; all Holm-adjusted p = 1.00). The largest Cox gain was +0.011 for BASE+HM (19 of 25 splits), and SHAP-selected features did not outperform all features. The logistic regression and LightGBM AUCs classify observed status and are descriptive.

Table 3. All patients (n = 1,295; 545 recurrences), 5 × 5 repeated CV, mean over 25 splits. AUC columns: descriptive classification of observed status (censoring ignored). Cox: fixed α = 10; Holm family = the six additions vs BASE.

Feature set	Features	LR AUC	LightGBM AUC	Cox C-index	Cox gain (splits improved)
BASE	21	0.659	0.636	0.649	—
BASE+DM	28	0.654	0.631	0.652	+0.003 (19/25)
BASE+HM	71	0.663	0.627	0.660	+0.011 (19/25)
BASE+PC	30	0.657	0.624	0.654	+0.005 (17/25)
BASE+DRIVERS	41	0.657	0.642	0.652	+0.003 (14/25)
BASE+GEN	98	0.655	0.636	0.653	+0.004 (15/25)
BASE+SHAPSEL	31	0.656	0.636	0.654	+0.005 (14/25)

3.4 ER-positive patients and the five-year landmark

In ER-positive patients, adding genomic blocks increased the fixed-penalty Cox C-index (Table 4): Hallmark scores by +0.034 (95% CI +0.007 to +0.062; 24 of 25 splits; Holm p = 0.086) and all genomic features by +0.032 (+0.000 to +0.064; 23 of 25 splits; Holm p = 0.19). Against the strictly clinical comparator the gain was +0.031 (Holm p = 0.19), and the exploratory CORE4 set added +0.015 (Holm p = 0.052). The evaluated genomic feature sets did not improve discrimination in the five-year landmark analysis (ER-positive −0.025, 95% CI −0.078 to +0.027; 188 events).

Table 4. Fixed-penalty Cox (α = 10), C-index, 5 × 5 repeated CV. Landmark: recurrence-free and under follow-up at 60 months, time from month 60. Holm family per analysis: the four blocks shown, CORE4, and CLIN+GEN vs CLIN.

Analysis	n (events)	BASE	+DM	+HM	+DRIVERS	+GEN
All patients	1,295 (545)	0.649	0.652	0.660	0.652	0.653
ER-positive	996 (408)	0.627	0.640	0.661	0.635	0.659
Landmark 5 y, all	873 (211)	0.621	0.613	0.601	0.614	0.592
Landmark 5 y, ER-positive	703 (188)	0.596	0.587	0.573	0.590	0.571

3.5 Features the binary classifier relied on

In the binary LightGBM classifier for all patients (Table 5), positive lymph nodes, tumor size, and NPI were the strongest features. Among molecular features, the copy-number fraction ranked in the top 15 in every split, followed by the reactive oxygen species pathway and the G2M checkpoint and mitotic spindle sets. When SHAP selected genomic features inside training folds, the copy-number fraction was chosen in 100% of splits, reactive oxygen species in 96%, G2M checkpoint in 76%, and mitotic spindle in 60%. These attributions describe this classifier only. In the ER-positive Cox analysis, the Hallmark block accounted for most of the gain (+0.034), whereas the 20 driver features, which include the copy-number fraction, added +0.008 (Table 4).

Table 5. TreeSHAP stability, binary LightGBM classifier on BASE+GEN, all patients, 25 test folds (log-odds scale).

Feature	Type	Mean |SHAP|	In top 15
Positive lymph nodes	clinical	0.321	100%
Copy-number fraction (assayed genes altered)	copy number	0.149	100%
Tumor size	clinical	0.138	100%
Nottingham Prognostic Index	clinical	0.137	96%
Reactive oxygen species pathway	expression	0.125	96%
G2M checkpoint	expression	0.102	80%
Mitotic spindle	expression	0.102	64%

3.6 Methylation and immune summaries

No improvement was demonstrated with the evaluated methylation and immune feature definitions and models (Table 6). Relative to BASE+GEN in the ER-positive methylation subset, adding methylation changed the C-index by −0.004 (95% CI −0.012 to +0.005), immune summaries by −0.004 (−0.011 to +0.003), and both by −0.008 (−0.019 to +0.004). The immune summaries were strongly correlated with existing gene-set scores (T-cell summary vs Desmedt immune module r = 0.83; fibroblasts vs Hallmark EMT r = 0.93). In the smaller methylation subset, the genomic gain over BASE in ER-positive patients was +0.020 (−0.013 to +0.053; 21 of 25 splits).

Table 6. Methylation and immune summaries: paired contrasts, fixed-penalty Cox (α = 10), 5 × 5 repeated CV. Holm families within each cohort: additions vs BASE, and additions vs BASE+GEN.

Comparison	Cohort (n; events)	ΔC-index (95% CI)	Splits improved	Holm p
+GEN+METH vs BASE+GEN	ER+, methylation subset (736; 307)	−0.004 (−0.012 to +0.005)	11/25	0.53
+GEN+IMM vs BASE+GEN	ER+, methylation subset (736; 307)	−0.004 (−0.011 to +0.003)	6/25	0.53
+GEN+METH+IMM vs BASE+GEN	ER+, methylation subset (736; 307)	−0.008 (−0.019 to +0.004)	5/25	0.53
+GEN+IMM vs BASE+GEN	ER+, full cohort (996; 408)	−0.006 (−0.014 to +0.002)	4/25	0.15
+METH vs BASE	ER+, methylation subset (736; 307)	+0.003 (−0.014 to +0.020)	17/25	1.00
+IMM vs BASE	ER+, full cohort (996; 408)	0.000 (−0.018 to +0.017)	15/25	0.96
+GEN vs BASE	ER+, methylation subset (736; 307)	+0.020 (−0.013 to +0.053)	21/25	0.68
+GEN+METH vs BASE+GEN	All, methylation subset (945; 404)	−0.002 (−0.014 to +0.010)	8/25	1.00

3.7 Tuned survival models

With tuning inside inner folds and 50 splits (Tables 7 and 8), the reduced-feature random survival forest increased the ER-positive C-index from 0.629 to 0.662 (+0.033, 95% CI +0.015 to +0.051; 48 of 50 splits; Holm p = 0.002), and the forest with all genomic features by +0.030 (Holm p = 0.011). Against the strictly clinical comparator, the forest gains were +0.040 and +0.043. Tuned Cox gains were +0.026 (95% CI −0.002 to +0.054;

Holm p = 0.28) with all genomic features and +0.013 with the reduced set. These are internal exploratory results. Uno's C-index at 120 months gave similar ER-positive increments (reduced forest +0.033, 95% CI +0.013 to +0.054; Cox with all genomic features +0.027, −0.001 to +0.055). For all patients, gains were +0.008 to +0.017 and not significant after Holm adjustment. The Cox penalty was mostly α = 100 in ER-positive patients (58–88% of fits) but α = 1000 in 74–86% of all-patient fits with genomic features, so a wider grid may matter; the forest chose the smaller leaf size (15) in 84–96% of ER-positive genomic fits.

Table 7. Tuned survival models, C-index, 10 × 5 repeated CV (50 splits).

Patients	Feature set	Features	Tuned Cox	Tuned RSF
ER-positive (n = 996)	BASE	19	0.632	0.629
ER-positive	BASE+GEN	96	0.658	0.659
ER-positive	BASE+PC+DRV	48	0.644	0.662
ER-positive	CLIN	17	0.632	0.620
ER-positive	CLIN+GEN	94	0.656	0.660
ER-positive	CLIN+PC+DRV	46	0.644	0.662
All (n = 1,295)	BASE	21	0.652	0.656
All	BASE+GEN	98	0.662	0.670
All	BASE+PC+DRV	50	0.660	0.668
All	CLIN	19	0.650	0.653
All	CLIN+GEN	96	0.662	0.669
All	CLIN+PC+DRV	48	0.660	0.668

Table 8. ER-positive gains over the matching baseline (tuned models, 50 splits). Holm family: the four comparisons within each model.

Comparison	Gain	95% CI	Splits improved	Holm p
Cox, BASE+GEN vs BASE	+0.026	−0.002 to +0.054	44/50	0.28
Cox, BASE+PC+DRV vs BASE	+0.013	−0.009 to +0.034	40/50	0.47
Cox, CLIN+GEN vs CLIN	+0.025	−0.003 to +0.052	43/50	0.28
Cox, CLIN+PC+DRV vs CLIN	+0.012	−0.011 to +0.035	38/50	0.47
RSF, BASE+GEN vs BASE	+0.030	+0.007 to +0.053	48/50	0.011
RSF, BASE+PC+DRV vs BASE	+0.033	+0.015 to +0.051	48/50	0.002
RSF, CLIN+GEN vs CLIN	+0.040	+0.015 to +0.065	49/50	0.004
RSF, CLIN+PC+DRV vs CLIN	+0.043	+0.024 to +0.061	49/50	<0.001

3.8 Exploratory assessment in TCGA-BRCA

After exclusions, 902 TCGA patients (100 events) had complete data, of whom 679 were ESR1-inferred ER-positive (66 events). Median follow-up estimated by reverse Kaplan–Meier was 29.9 months overall and 28.4 months in the ESR1-inferred ER-positive subgroup. In this harmonized TCGA assessment, the Cox model retained a small positive point estimate of incremental discrimination (+0.014, 95% CI −0.036 to +0.064), whereas the full RSF did not (−0.029); all gain intervals included zero (Table 9). The estimated external gain is imprecise and was not robust to censoring weighting: Uno's C-index at 36 months gave −0.009 (−0.067 to +0.045) for Cox. Excluding the mutation count, which is not comparable between panel and exome sequencing,

gave a Cox gain of +0.016 (−0.032 to +0.064). Inside METABRIC, the same harmonized comparison (Cox, 5 × 5 CV) gave an ER-positive gain of +0.040 (95% CI +0.011 to +0.070; 23 of 25 splits; raw p = 0.009).

Table 9. Exploratory assessment: models refitted on METABRIC with harmonized predictors and applied to TCGA-BRCA. C-index with 1,000 paired bootstrap resamples.

Subgroup (TCGA test set)	Model	Harmonized clinical	+ Genomic	Gain (95% CI)	Resamples with a gain
ESR1-inferred ER+ (679; 66 events)	Cox	0.608	0.623	+0.014 (−0.036 to +0.064)	74%
ESR1-inferred ER+	Cox, reduced	0.608	0.615	+0.007 (−0.038 to +0.048)	64%
ESR1-inferred ER+	RSF	0.606	0.577	−0.029 (−0.117 to +0.054)	26%
ESR1-inferred ER+	RSF, reduced	0.606	0.594	−0.012 (−0.090 to +0.066)	38%
All (902; 100 events)	Cox	0.649	0.653	+0.005 (−0.031 to +0.041)	59%
All	RSF	0.634	0.613	−0.021 (−0.069 to +0.027)	22%

Reduced = 9 principal components (PCA fitted on METABRIC) + 19 driver features. HCLIN = age, T category, N category, ER status (TCGA: ESR1-inferred), ERBB2 amplification, radiotherapy; ER status is constant in the ER-positive subgroup and was dropped.

4. Discussion

4.1 Principal findings

1. **An internal signal in ER-positive disease.** The clearest internal signal was observed in the ER-positive subgroup: adding genomic features increased discrimination by about 0.02–0.04 across fixed and tuned Cox models and random survival forests, with the strongest adjusted result for the reduced-feature forest. These results come from adaptive model development on one cohort and are exploratory; favorable repeated splits are not independent replications. Commercial multigene signatures also add prognostic information mainly in ER-positive disease [4,5], but differences in populations, endpoints, treatment, and follow-up preclude comparing the size of the gains.
2. **All patients combined.** Improvement was not demonstrated for all patients combined. We did not report an ER-negative subgroup comparison, so these results do not establish whether genomic features add value in ER-negative disease or whether the effect differs by ER status. Benchmark studies similarly find that multi-omic models rarely beat a clinical Cox model [11,12].
3. **Five-year landmark.** The evaluated genomic feature sets did not improve discrimination in the five-year landmark analysis (ER-positive −0.025, 95% CI −0.078 to +0.027). This does not establish that genomic information cannot predict late recurrence: the analysis had 188 ER-positive events, and Rueda et al. used a multistate model with distant-relapse endpoints [9], which our endpoint does not reproduce. Given the persistent late risk in ER-positive disease [6,7], larger cohorts or a small pre-specified feature set are needed.
4. **Which features carried the signal.** In the ER-positive Cox analysis, most of the gain came from the Hallmark expression block; in the binary classifier for all patients, the copy-number fraction and proliferation and oxidative-stress gene sets were the most stable attributions. These are model-dependent predictive attributions (Section 4.2).

5. **Methylation and immune summaries.** No improvement over the genomic model was demonstrated with the evaluated feature definitions and models; the immune summaries were strongly correlated with existing gene-set scores.
6. **Transport.** In this harmonized TCGA assessment, the Cox model retained a small positive point estimate of incremental discrimination, whereas the full RSF did not; all gain intervals included zero. This does not establish that one algorithm family transfers better: no direct paired comparison between algorithms was made, and the comparator, subgroup definition, and endpoint also changed.
7. **Context.** The DeepSurv benchmark on METABRIC reported C-indices of 0.62–0.65 for Cox, DeepSurv, and RSF on overall survival with nine clinical and expression features [39], and multimodal models that add digital pathology reached a pooled C-index of about 0.71 across large external cohorts [13]. These studies differ in endpoints and populations and are context, not benchmarks for our estimates. Recurrence also depends on factors not measured at diagnosis, such as treatment adherence.

4.2 Biological interpretation

Copy-number fraction. The most stable molecular attribution in the binary classifier was the fraction of assayed genes with a non-zero copy-number call, a gene-count proxy for copy-number instability rather than a length-weighted fraction of the genome. Chromosomal instability generates the copy-number diversity on which selection acts during dissemination and therapy. In METABRIC, the ER-positive Integrative Clusters with the highest late-relapse risk (IntClust 1, 2, 6, and 9) are defined by focal amplifications [8,9], and whole-genome sequencing has shown that high-risk ER-positive tumors share a structural archetype with HER2-positive tumors, marked by complex focal and extrachromosomal amplifications, whereas typical-risk ER-positive tumors are largely genomically stable [10]. A single genome-wide gene count is a coarse readout of that biology, and in the ER-positive Cox analysis the driver block containing it added little (+0.008). Copy-number and expression models have also recently been linked to long-term recurrence-free survival in METABRIC [40].

Proliferation. AURKA, E2F targets, and G2M checkpoint belonged to the same correlation cluster; the mitotic spindle score formed a separate cluster (Figure 2). Proliferation is the core of commercial signatures that add to clinical variables in ER-positive disease: the 21-gene Recurrence Score includes a proliferation group (Ki-67, AURKA, BIRC5, CCNB1, MYBL2) [2], the PAM50 risk-of-recurrence score includes a proliferation index [3,41], and proliferation is the dominant prognostic module in ER+/HER2− tumors [5]. Histologic grade captures proliferation only coarsely, and nearly half of our ER-positive patients had grade 2 tumors (Table 2), where continuous proliferation scores may help distinguish tumors with different outcomes. This is a hypothesis for why the internal signal was clearest in ER-positive disease; it was not tested directly.

Oxidative stress. The reactive oxygen species pathway was the most stable expression attribution in the binary classifier. This Hallmark set mainly contains antioxidant and redox-buffering genes and is correlated with the metabolic cluster rather than with proliferation. Redox adaptation could plausibly support survival of disseminated cells and resistance to therapy, but this finding was not anticipated and is hypothesis-generating.

4.3 Clinical utility

This study shows a statistical improvement in internal discrimination, not clinical usefulness. The observed increase concerns concordance among comparable patient pairs; it does not quantify the proportion of patients whose treatment recommendation would change. Whether the models would change decisions, such as extending endocrine therapy beyond five years [7] or offering chemotherapy, depends on calibrated risks

and clinically meaningful thresholds. We did not assess calibration, reclassification, decision curves [42], or proportional-hazards and functional-form assumptions, so the paper makes discrimination claims only. A clinically oriented evaluation would define the decision and thresholds, report calibration (calibration slope and Brier score) at horizons supported by follow-up, and benchmark against an established signature such as the Recurrence Score or PAM50 risk of recurrence in the same patients.

4.4 Limitations

- **Retrospective cohorts.** METABRIC patients were diagnosed in 1977–2005, before routine trastuzumab and modern endocrine strategies.
- **External assessment.** TCGA has short follow-up (median 28.4 months by reverse Kaplan–Meier in the ESR1-inferred ER-positive subgroup; 66 events), a different endpoint, different platforms, harmonized proxies that omit grade and NPI, ESR1-inferred ER status, and molecular standardization estimated on the unlabeled TCGA cohort. It refits a separately specified harmonized model and does not evaluate a frozen predictor for a single new patient.
- **Upstream normalization.** Expression z-scores were normalized by the data provider against a cohort reference before our pipeline, and this step was not repeated inside folds; a strictly inductive analysis that re-estimates the reference within each training partition, and its sensitivity analysis, were not performed.
- **Complete-case analysis.** The modelling cohort (1,295 of 2,509; 51.6%) differs from excluded patients with a recorded outcome in recurrence, receptor status, grade, nodal status, and treatment (Section 2.1), so results may not apply to the full cohort.
- **Adaptive analysis and multiplicity.** The reduced representation, tuned models, strictly clinical comparator, and CORE4 were introduced after earlier analyses of the same data, CORE4 after inspecting SHAP results. Holm adjustment covers each declared comparison family, not all adaptive decisions, so the significant reduced-forest result (Holm $p = 0.002$) is an internal exploratory finding.
- **Discrimination only.** Calibration, Brier score, and decision analysis were not assessed. Harrell's C-index depends on the censoring distribution [43]; the IPCW check agreed internally but changed the sign of the external Cox gain.
- **Endpoint.** Relapse-free status combines local and distant relapse and does not model competing risks.
- **Methylation.** The cBioPortal file has no normal-tissue reference, so instability measures are within-tumor approximations of the published definitions.
- **Tuning and treatment variables.** Tuning grids were small, and the Cox penalty often selected the upper boundary in all patients. Treatment variables are known only after treatment planning and should not be interpreted causally.

4.5 Future work

1. A deep survival model (for example DeepSurv [39]) with careful tuning, compared with the tuned Cox and random survival forest.
2. A wider penalty grid for Cox and elastic-net Cox as an alternative.
3. Calibration of the survival models (calibration slope and Brier scores at horizons supported by follow-up), and decision-curve analysis once clinical decisions and thresholds are defined [42].

4. A strictly inductive pipeline that re-estimates expression normalization within training data, and a frozen preprocessing and prediction pipeline evaluated in an independent cohort with long follow-up.
5. Paired block-ablation experiments in the ER-positive survival models, and a segment-length-weighted copy-number instability measure.
6. Spatial and imaging data, such as METABRIC imaging mass cytometry [44] or digital pathology [13], which capture tumor architecture and immune spatial organization not captured by bulk expression.

5. Conclusion

In this exploratory analysis, genomic summaries improved internal discrimination in the ER-positive METABRIC subgroup, with the strongest adjusted result observed for a reduced-feature random survival forest. The harmonized TCGA analysis yielded an imprecise positive gain for Cox and did not confirm improved external performance. The findings support further evaluation of molecular information for recurrence risk stratification, but do not establish clinical utility or equivalence to commercial prognostic assays.

Data and code availability

METABRIC and TCGA-BRCA data are publicly available from cBioPortal (https://www.cbioportal.org/datasets: *brca_metabric* and *brca_tcga_pan_can_atlas_2018*). Gene sets came from MSigDB (human Hallmark collection, release 2026.1, h.all.v2026.1.Hs), the genefu package (Desmedt modules), and MCPcounter (marker genes). All code, the processed dataset, the feature dictionary, and every result table are available at https://github.com/trinhamcity1/METABRIC_Recurrence_Multiomics; *run_all.sh* regenerates every table and figure in this paper from the processed data (docs/TABLE_FIGURE_MANIFEST.md maps each element to its script and output file). Results were produced with Python 3.10.12 on Linux (numpy 2.2.6, pandas 2.3.3, scikit-learn 1.7.2, scikit-survival 0.25.0, LightGBM 4.7.0, SHAP 0.49.1), pinned in requirements-lock.txt; fitted TCGA models, scaling and PCA objects, and out-of-fold predictions are saved in the repository.

Ethics and data use

This secondary analysis used publicly available, de-identified data from cBioPortal, collected under the consent and ethics approvals of the original METABRIC and TCGA studies. No new data were collected and no additional ethics approval was sought.

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Conflicts of interest

T.N. declares no conflicts of interest. [Other authors to confirm, including any interests related to the KlinicSync affiliation.]

Author contributions

T.N.: conceptualization, data curation, software, formal analysis, investigation, visualization, writing – original draft, and writing – review and editing. P.G.-R.: conceptualization, supervision, project administration, writing – original draft, and writing – review and editing. H.B.A.-A.: methodology, validation (review of the analysis and code repository), and writing – review and editing. M.A.B.-O.: methodology and writing – review and editing. R.T.-S.: methodology, supervision, and writing – review and editing. A.H. and G.A.: validation (clinical interpretation of the findings) and writing – review and editing. All authors read and approved the final manuscript.

References

1.Nham T, Arteaga-Arteaga HB, Bravo-Ortiz MA, Tabares-Soto R, Hernandez A, Arenas G, Guillen-Rondon P. Multi-target machine learning for breast cancer prognosis and molecular classification using METABRIC. In: *Proceedings of EGETC 2026*. 2026.

2.Paik S, Shak S, Tang G, et al. A multigene assay to predict recurrence of tamoxifen-treated, node-negative breast cancer. *N Engl J Med* 2004;351:2817–2826.

3.Dowsett M, Sestak I, Lopez-Knowles E, et al. Comparison of PAM50 risk of recurrence score with Oncotype DX and IHC4 for predicting risk of distant recurrence after endocrine therapy. *J Clin Oncol* 2013;31:2783–2790.

4.Sestak I, Buus R, Cuzick J, et al. Comparison of the performance of 6 prognostic signatures for estrogen receptor-positive breast cancer: a secondary analysis of a randomized clinical trial. *JAMA Oncol* 2018;4:545–553.

5.Desmedt C, Haibe-Kains B, Wirapati P, et al. Biological processes associated with breast cancer clinical outcome depend on the molecular subtypes. *Clin Cancer Res* 2008;14:5158–5165.

6.Pan H, Gray R, Braybrooke J, et al. 20-year risks of breast-cancer recurrence after stopping endocrine therapy at 5 years. *N Engl J Med* 2017;377:1836–1846.

7.Early Breast Cancer Trialists' Collaborative Group (EBCTCG). Extending the duration of endocrine treatment for early breast cancer: patient-level meta-analysis of 12 randomized trials of aromatase inhibitors in 22 031 postmenopausal women already treated with at least 5 years of endocrine therapy. *Lancet* 2025;406:603–614.

8.Curtis C, Shah SP, Chin SF, et al. The genomic and transcriptomic architecture of 2,000 breast tumours reveals novel subgroups. *Nature* 2012;486:346–352.

9.Rueda OM, Sammut SJ, Seoane JA, et al. Dynamics of breast-cancer relapse reveal late-recurring ER-positive genomic subgroups. *Nature* 2019;567:399–404.

10.Houlahan KE, Mangiante L, Sotomayor-Vivas C, et al. Complex rearrangements fuel ER+ and HER2+ breast tumors. *Nature* 2025;638:510–518.

11.Herrmann M, Probst P, Hornung R, Jurinovic V, Boulesteix AL. Large-scale benchmark study of survival prediction methods using multi-omics data. *Brief Bioinform* 2021;22:bbaa167.

12.Wissel D, Rowson D, Boeva V. Systematic comparison of multi-omics survival models reveals a widespread lack of noise resistance. *Cell Rep Methods* 2023;3:100461.

13.Witowski J, Zeng KG, Cappadona J, et al. Multi-modal AI for comprehensive breast cancer prognostication. Nat Commun 2026;17:5879. doi:10.1038/s41467-026-73088-y.

14.Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ* 2024;385:e078378.

15.Cerami E, Gao J, Dogrusoz U, et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov* 2012;2:401–404.

16.Gao J, Aksoy BA, Dogrusoz U, et al. Integrative analysis of complex cancer genomics and clinical profiles using the cBioPortal. *Sci Signal* 2013;6:pl1.

17.de Bruijn I, Kundra R, Mastrogiacomo B, et al. Analysis and visualization of longitudinal genomic and clinical data from the AACR Project GENIE Biopharma Collaborative in cBioPortal. *Cancer Res* 2023;83:3861–3867.

18.Pereira B, Chin SF, Rueda OM, et al. The somatic mutation profiles of 2,433 breast cancers refine their genomic and transcriptomic landscapes. *Nat Commun* 2016;7:11479.

19.Liberzon A, Birger C, Thorvaldsdóttir H, Ghandi M, Mesirov JP, Tamayo P. The Molecular Signatures Database hallmark gene set collection. *Cell Syst* 2015;1:417–425.

20.Castanza AS, Recla JM, Eby D, Thorvaldsdóttir H, Bult CJ, Mesirov JP. Extending support for mouse data in the Molecular Signatures Database (MSigDB). *Nat Methods* 2023;20:1619–1620.

21.Batra RN, Lifshitz A, Vidakovic AT, et al. DNA methylation landscapes of 1538 breast cancers reveal a replication-linked clock, epigenomic instability and cis-regulation. *Nat Commun* 2021;12:5406.

22.Ali HR, Chlon L, Pharoah PDP, Markowetz F, Caldas C. Patterns of immune infiltration in breast cancer and their clinical implications: a gene-expression-based retrospective study. *PLoS Med* 2016;13:e1002194.

23.Loi S, Drubay D, Adams S, et al. Tumor-infiltrating lymphocytes and prognosis: a pooled individual patient analysis of early-stage triple-negative breast cancers. *J Clin Oncol* 2019;37:559–569.

24.Becht E, Giraldo NA, Lacroix L, et al. Estimating the population abundance of tissue-infiltrating immune and stromal cell populations using gene expression. *Genome Biol* 2016;17:218.

25.Harrell FE Jr, Lee KL, Mark DB. Multivariable prognostic models: issues in developing models, evaluating assumptions and adequacy, and measuring and reducing errors. *Stat Med* 1996;15:361–387.

26.Pölsterl S. scikit-survival: a library for time-to-event analysis built on top of scikit-learn. *J Mach Learn Res* 2020;21(212):1–6.

27.Ishwaran H, Kogalur UB, Blackstone EH, Lauer MS. Random survival forests. *Ann Appl Stat* 2008;2:841–860.

28.Ke G, Meng Q, Finley T, et al. LightGBM: a highly efficient gradient boosting decision tree. *Adv Neural Inf Process Syst* 2017;30:3146–3154.

29.Wang C, Wu Q, Weimer M, Zhu E. FLAML: a fast and lightweight AutoML library. *Proc Mach Learn Syst* 2021;3:434–447.

30.Płońska A, Płoński P. MLJAR: state-of-the-art automated machine learning framework for tabular data (mljar-supervised). 2021. https://github.com/mljar/mljar-supervised

31.Bouckaert RR, Frank E. Evaluating the replicability of significance tests for comparing learning algorithms. In: *Advances in Knowledge Discovery and Data Mining (PAKDD 2004)*. LNCS 3056. Springer; 2004:3–12.

32.Nadeau C, Bengio Y. Inference for the generalization error. *Mach Learn* 2003;52:239–281.

33.Holm S. A simple sequentially rejective multiple test procedure. *Scand J Stat* 1979;6:65–70.

34.Bates S, Hastie T, Tibshirani R. Cross-validation: what does it estimate and how well does it do it? *J Am Stat Assoc* 2024;119:1434–1445.

35.Lundberg SM, Erion G, Chen H, et al. From local explanations to global understanding with explainable AI for trees. *Nat Mach Intell* 2020;2:56–67.

36.Cancer Genome Atlas Network. Comprehensive molecular portraits of human breast tumors. *Nature* 2012;490:61–70.

37.Collins GS, Dhiman P, Ma J, et al. Evaluation of clinical prediction models (part 1): from development to external validation. *BMJ* 2024;384:e074819.

38.Liu J, Lichtenberg T, Hoadley KA, et al. An integrated TCGA pan-cancer clinical data resource to drive high-quality survival outcome analytics. *Cell* 2018;173:400–416.e11.

39.Katzman JL, Shaham U, Cloninger A, Bates J, Jiang T, Kluger Y. DeepSurv: personalized treatment recommender system using a Cox proportional hazards deep neural network. *BMC Med Res Methodol* 2018;18:24.

40.Rotbauer M, Dawe M, Nixon KCJ, et al. The role of circulating tumor cell-associated genes in the progression of estrogen receptor-positive breast cancer. npj Breast Cancer 2026;12:11 (published online 15 December 2025). doi:10.1038/s41523-025-00874-0.

41.Parker JS, Mullins M, Cheang MCU, et al. Supervised risk predictor of breast cancer based on intrinsic subtypes. *J Clin Oncol* 2009;27:1160–1167.

42.Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Med Decis Making* 2006;26:565–574.

43.Uno H, Cai T, Pencina MJ, D'Agostino RB, Wei LJ. On the C-statistics for evaluating overall adequacy of risk prediction procedures with censored survival data. *Stat Med* 2011;30:1105–1117.

44.Danenberg E, Bardwell H, Zanotelli VRT, et al. Breast tumor microenvironment structures are associated with genomic features and clinical outcome. *Nat Genet* 2022;54:660–669.